

A binomial experiment

A binomial experiment has a fixed number of independent trials, two outcomes per trial, and constant success probability p .

Do ten independent fair-coin tosses satisfy these conditions?

Bernoulli trials

A Bernoulli trial has two outcomes called success and failure. A binomial experiment counts successes in n independent Bernoulli trials with the same p .

What is one Bernoulli trial in a transmission experiment?

Define success

Each independent transmission either succeeds or fails.

Must “success” mean a desirable outcome in a binomial model?

Define the random variable

Ten independent transmissions each succeed with probability 0.8.

What does X count in a binomial model?

Identify n and p

Twenty independent transmissions each succeed with probability 0.9.

What are n and p ?

Success and failure probabilities

The probability of failure is $q = 1 - p$. Each independent transmission succeeds with probability $p = 0.85$.

What is q , and which outcome does it describe?

Possible values

X counts successes in 6 independent trials.

What values can X take?

Binomial notation

The binomial probability distribution uses n trials and success probability p .

$$X \sim B(n, p)$$

What experiment could be described by $X \sim B(8, 0.25)$?

Translate exactly

Ten independent transmissions each succeed with probability 0.8. X counts successes.

How would you write “exactly eight successes” as a probability?

Translate at least

Ten independent transmissions each succeed with probability 0.8. X counts successes.

How would you write “at least eight successes”?

Translate fewer than

Ten independent transmissions each succeed with probability 0.8. X counts successes.

Which values contribute to $P(X < 3)$?

One ordered sequence

Four independent trials each succeed with probability 0.3.

What is the probability of the ordered sequence success, failure, success, failure?

Count the arrangements

Four independent trials produce exactly two successes.

How many different placements of the two successes are possible?

The binomial probability

The binomial formula combines the number of arrangements with each arrangement's probability.

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n - x}$$

For $n = 4$ and $p = 0.3$, what is $P(X = 2)$?

A full binomial distribution

Let X count successes in three independent trials, each with $p = 0.25$.

What probability table and probability bar graph describe X ?

No successes

Five independent trials each succeed with probability 0.2.

What is $P(X = 0)$?

At least one success

Five independent trials each succeed with probability 0.2.

How can a complement simplify $P(X \geq 1)$?

A cumulative probability

Five independent trials each succeed with probability 0.2.

Which exact probabilities add to $P(X \leq 2)$?

Technology result

For $X \sim B(10, 0.8)$, software reports $P(X \leq 7) = 0.3222$.

What is $P(X \geq 8)$?

PDF and CDF commands

For a discrete model, PDF gives $P(X = x)$; the cumulative distribution function (CDF) gives $P(X \leq x)$.

For $X \sim B(10, 0.8)$, which command evaluates exactly eight successes and which evaluates at most eight?

Expected successes

A binomial count has mean np .

$$\mu = np$$

For 20 independent trials with $p = 0.3$, what is the expected number of successes?

Standard deviation

A binomial count has standard deviation $\sqrt{np(1 - p)}$.

$$\sigma = \sqrt{np(1 - p)}$$

For $n = 20$ and $p = 0.3$, what is σ ?

Binomial variance

For a binomial count, variance is $\sigma^2 = npq$ and standard deviation is its square root.

For $n = 20$, $p = 0.3$, $q = 0.7$, how do the variance and standard deviation differ?

A fixed count?

A binomial model has $n = 20$ and $p = 0.3$.

Does an expected count of 6 guarantee exactly 6 successes?

Sampling without replacement

A box has 3 red and 2 blue tokens. Draw 4 uniformly without replacement; X counts red tokens.

Which binomial condition fails?

Changing probabilities

A machine's success probability declines as its battery runs down.

Which binomial condition needs reconsideration?

Toward a continuous approximation

A binomial model has $n = 100$ and $p = 0.4$.

How many successes and failures are expected before considering a later normal approximation?